MAINS MASTER PROGRAM (MMP) 2024
DISASTER MANAGEMENT-1

1. TABLE OF CONTENTS

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

2. SYLLABUS: DISASTER AND DISASTER MANAGEMENT

3. PYQS (2013- 2023)

i. How important are **vulnerability and risk assessment** for pre-disaster management? As an administrator, what are key areas that you would focus on in a Disaster Management System? [2013] [10 marks]

ii. **Drought** has been recognized as a disaster in view of its spatial expanse, temporal duration, slow onset and lasting effects on vulnerable sections. With a focus on the September **2010 guidelines from the National Disaster Management Authority (NDMA)**, discuss the mechanisms for preparedness to deal with likely El Nino and La Nina fallouts in India. [2014, 12.5 marks]

iii. The frequency of **earthquakes** appears to have increased in the Indian subcontinent. However, **India's preparedness** for mitigating their impact has significant gaps. Discuss various aspects [2015, 12.5 marks]

iv. The frequency of **urban floods** due to high intensity rainfall is increasing over the years. Discussing the reasons for urban floods, highlight the mechanisms for preparedness to reduce the risk during such events [2016, 12.5 marks]

v. With reference to **National Disaster Management Authority (NDMA) guidelines**, discuss the measures to be adopted to mitigate the impact of recent incidents of **cloudbursts** in many places of Uttarakhand. [2016, 12.5 marks]

vi. On December 2004, **Tsunami** brought havoc on 14 countries including India. Discuss the factors responsible for occurrence of Tsunami and its effects on life and economy. In the **light of guidelines of NDMA (2010)** describe the mechanisms for preparedness to reduce the risk during such event. 2017, 15 marks]

vii. Describe various **measures taken in India for Disaster Risk Reduction (DRR)** before and after signing '**Sendai Framework for DRR (2015-2030)'**. How is this framework different from '**Hyogo Framework for Action, 2005'**? (2018, 250 Words, 15 Marks)

viii. **Vulnerability** is an essential element for defining disaster impacts and its threat to people. How and in what ways can vulnerability to disasters be characterized? Discuss different types of vulnerability with reference to disasters [2019, 10 marks, 150 words]

ix. **Disaster preparednes**s is the first step in any disaster management process. Explain how hazard zonation mapping will help in disaster mitigation in the case of **landslides** [2019, 15 marks, 250 words]

x. Discuss the recent measures initiated in disaster management by the Government of India departing from the earlier reactive approach [2020, 15 marks, 250 words]

xi. Discuss about the vulnerability of India to **earthquake** related hazards. Give examples including the salient features of major disasters caused by earthquakes in different parts of India during the last three decades. (2021, 10 marks, 150 words)

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

xii. Describe the various causes and the effects of l**andslides**. Mention the important components of the National Landslide Risk Management Strategy. (2021, 10 marks, 250 words)

xiii. Explain the mechanism and occurrence of **<u>cloudburst</u>** in the context of the Indian subcontinent. Discuss two recent examples. [2022, 10 marks, 150 words]

xiv. **<u>Dam Failures</u>** are always catastrophic, especially on the downstream side, resulting in colossal loss of life and property. Analyze the various causes of Dam failures. Give two example of large dam failures. [2023, 10 marks, 150 words]

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

4. BASICS OF DISASTER AND DISASTER MANAGEMENT

- **"Disaster"** is defined under section 2(d) of the Disaster Management Act, 2005 as a **catastrophe, mishap, calamity or grave occurrence** in any area, arising from **natural or man-made causes**, and is of such nature or magnitude as to be **beyond the coping capacity of the affected area**.

- Disasters, whether natural or man-made, have been part of man's evolution since times immemorial. Many civilizations including the ancient Indus Valley civilization is thought to have declined because of some natural or man-made disasters.

1) KEY COMPONENTS OF DISASTER MANAGEMENT

- A disaster needs to be examined in terms of its **management cycle** that would enable us to anticipate the crisis, prevent and mitigate it to the extent possible and deal with the crisis situation as it emerges.

- The life cycle of disaster Management can be divided into **three phases:**

i. Pre-Crisis: Risk Reduction and Preparedness

- This is the period when the potential hazard risk and vulnerabilities can be assessed and steps taken for preventing and mitigating the crisis and preparing for actual occurrence.
- **It includes:**
  - Creation of legal and institutional framework
  - Hazard Mapping and Vulnerability Analysis.
  - Adopting risk reduction techniques
  - Infrastructure improvement for risk reduction
    - **Infra improvement can be both long term and short term.**
      a. **Long term Measures** - Embankments (floods); Augmenting Irrigation facilities, watershed management (drought proofing); afforestation (landslides); earthquake resistant structures and sound environmental practices.
      b. **Short term measures** may also reduce or modify the scale and intensity of the threat. This may include implementation of building codes; zoning regulations; maintenance of drainage system; improved awareness in public about disaster etc.
  - Setting up Early Warning Systems (EWS)
  - Awareness Generation
  - **Capacity building** of government institutions and agencies to ensure fast response during disaster.
  - Research and Development for more detailed understanding

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

ii. **During Crisis: Emergency Response**

- Disaster Response aims to provide **immediate attention** to maintain life, improve health, and to support the morale of the victim population.
  - It includes activities like **warning, search, rescue, evacuation,** followed by **provisions of basic needs** like first aid, medicine, food, clothing, shelter and other necessities essential to bring life of the affected back to a degree of normalcy.

iii. **Post Crisis: The Three Rs (Recovery Rehabilitation and Resettlement)**

- Post crisis activities can be summed in 3 R's: Recovery, Rehabilitation and Resettlement
  a. **Early Recovery**: This is the stage when efforts are made to achieve early recovery and reduce vulnerability and future risks. It comprises activities that encompass two **overlapping phases of rehabilitation and reconstruction**.
  b. **Rehabilitation** refers to actions taken in the aftermath of a disaster to enable basic services to resume functioning, assist victim's self help efforts to repair dwellings and community facilities, and to facilitate the revival of economic activities with more sustainable livelihoods.
  c. **Reconstruction** refers to **permanent construction** or replacement of severely damaged physical structures, the **full restoration of all services and local infrastructure**, and the revitalization of the economy (including agriculture). It should include **development of disaster resilient infrastructure** and must be fully integrated into long-term development plans.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

5. DISASTER SITUATION IN INDIA

- **Practice Question:**
  - » Discuss the key factors making India highly vulnerable to various disasters [10 marks, 150 words]
- India has recorded the **third highest number of natural disasters** (after China and USA) in last 20 years (2000-2019): Report "**Human Cost of Disasters**" by UN Office for Disaster Risk Reduction (UNDRR).
- **Vulnerability Profile of India:** India faces very high vulnerability due to various factors:
  - » **Adverse geo-climatic conditions**
    - ▪ **The geo tectonic features of the Himalayan regio**n and adjacent alluvial plains make the region susceptible to earthquakes, landslides, water erosion etc.
      - ▪ **58.6 percent of landmass** is prone to **earthquake** of moderate to high intensity.
      - ▪ **Hilly areas** are at risk from **landslides and avalanches**.
    - ▪ Of close to 7516 km **long coastline**, **close to 5700 km is prone to cyclones and Tsunamis**;
    - ▪ **Droughts:** The Western Parts of the country, including Rajasthan, Gujarat and some parts of Maharashtra are hit very frequently by **drought situation**. If Monsoon worsens the situation spreads to other parts of the country as well.

  - » **Climate Change:** It is expected to further increase the **frequency and intensity of current extreme weather events** and give rise to vulnerabilities with differential spatial and socio economic impacts on communities.
    - ▪ Sea-level rise -> coastal flooding
    - ▪ Melting of glaciers -> River overflow and flooding
    - ▪ Higher temperature -> Heat waves
    - ▪ Variation in weather pattern -> Floods and Droughts

  - » **Environmental Factors**
    - ▪ Increased land-degradation -> Famines
    - ▪ Water Pollution and unsustainable ground water extraction -> Drought
  - » **Socio-Economic Factors**
    - ▪ **High levels of poverty and risk to disasters** are inextricably linked and mutually reinforcing.
      - ▪ **Development within high risk zones**
      - ▪ For e.g. poverty forces people to live in disaster prone regions. The quality of infrastructure, houses etc. available to them are poor.
    - ▪ **High population density** -> easy spread of pandemic; More pressure on infrastructure - Road/Railway accidents etc.
    - ▪ **Unplanned Urbanization and unscientific development** -> Urban Floods, Building fires etc.
    - ▪ **Unregulated industrialization** -> River Pollution; Chemical accidents

- **Conclusion**:
  - » By adopting holistic approach which includes strengthening legal and institutional framework, hazard mapping, vulnerability analysis, improved infrastructure, establishment of early warning system, community preparedness etc. India can reduce the risk of these disasters and create a safer environment for citizens.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

6. WORLD CONFERENCE ON DISASTER RISK REDUCTION AND SENDAI FRAMEWORK

- **Introduction**: WCDRR is a series of UN conferences focused on disaster and climate risk management. They bring together various stakeholders including Government officials, Civil Society Organization, Local Bodies, Private Sector participant etc to discuss the disaster issues.
    - » So far, three conferences have been convened - Yokohama (1994); Kobe (2005) and Sendai (2015) - with following outcomes.

Conference	Outcome
First (1994)	Yokohama strategy and Plan of action of a safer world
Second (2005)	Hyogo Framework for Action 2005-2015 : Building the Resilience of Nations and Communities to Disasters
Third (2015)	Sendai Framework for Disaster Risk Reduction 2015-2030

- **Sendai Framework for Disaster Risk Reduction 2015-2030** is a **15 year non-binding agreement** which recognizes that the **state has the primary role t**o reduce disaster risk but that responsibility should be shared with other stakeholders including local government and the private sector.
    - » **Aim**: "The **substantial reduction of disaster risk and losses** lives, livelihood and health and in the economic, physical, social, cultural and environmental assets of persons, businesses, communities and countries.
    - » The framework has **four specific priorities for action**:
        - ▫ **Understanding disaster risk**
        - ▫ Strengthening disaster risk **governance** to manage disaster risk
        - ▫ **Investing** in disaster risk reduction for resilience
        - ▫ Enhancing disaster **preparedness** for effective response, and to "Build Back Better" in recovery, rehabilitation and reconstruction.

The Seven Global Targets

- **How is Sendai Framework Different from Hyogo Framework for Action?**
    - » **Scope:** While Hyogo primarily focused on natural hazards, the Sendai framework has expanded the scope to cover both natural and manmade hazards including technological hazards including chemical/industrial hazards, radiological, biological hazards etc.
    - » **Priorities**: Sendai emphasizes on disaster risk governance and investment which was missing in Hyogo framework.
    - » **Targets**: Hyogo had not set specific targets; whereas Sendai has led down 7 specific global targets.
    - » **Stakeholders**: While Hyogo emphasized on the role of government in disaster management, Sendai focuses on multi-stakeholder approach which includes government, local bodies, Civil Society, and private sector. It recognizes that effective disaster reductio requires collaboration, knowledge sharing, and participation from diverse sectors.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

» **Time Frame**: Hyogo Framework for Action had 10 years of implementation period; whereas Sendai Framework has 15 years of implementation period (2015-2030)

- **India has taken several steps for disaster risk reduction both before and after signing the Sendai Framework for Disaster Risk Reduction (2015-2030)**
  - **Pre Sendai Steps:**
    1. **Disaster Management Act, 2005** was enacted to provide legal framework to disaster risk reduction in India. It established various institutions and elaborated on their roles and responsibilities.
    2. **National Disaster Management Authority** (NDMA) was established in 2005 as the apex body for disaster management in India. It is responsible for policy formulation, coordination and implementation of DRR measures across the country.
    3. **National Disaster Response Force (NDRF)** was established in 2006 as a specialized force for disaster response. It consists of personnel trained in various aspects of disaster management and plays crucial role in rescue, relief, and response operations.
    4. **National Policy on Disaster Management**, 2009: It outlines the framework for disaster management in India and emphasizes on risk assessment, capacity development, and involvement of multiple stakeholders.

  - **Post Sendai Initiatives taken by India and the way forward**
    - In furtherance to its commitment to the Sendai framework, Government has taken up several important initiatives post Sendai Declaration.
      1. GoI has issued a set of priority actions to all the state governments based on the goals, targets and priorities of Sendai Framework.
      2. **National Disaster Response Force (NDRF) has been strengthened** , both in terms of state of art training and equipment to further empower the professional disaster response force.
         - GoI has also approved creation of **National Disaster Response Reserve (NDRR)** through a revolving fund of Rs 250 crore to be operated by NDRF. This dedicated fund will enable the NDRF to maintain a ready inventory of emergency goods and services comprising of tents, medicines, food items etc.
      3. Government has expressed its keenness to share India's expertise and help other countries in disaster response as it did during the Japan Earthquake in 2011 or the Nepal earthquake in 2015.
         - The government hosts SAARC Disaster Management Centre to reduce disaster risks in the region promoting knowledge sharing among the SAARC countries.
         - Similarly, INCOIS (Indian National Centre for Ocean Information Services) INCOIS, provides early warning not only to India but also to 28 countries in the Indian Ocean Rim.
         - India has hosted AMCDRR in Nov 2016 and adopted **'New Delhi Declaration' and 'Regional Action Plan for implementation of Sendai Framework'**.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

iv. NIDM has also signed an MoU with Jawaharlal Nehru University for establishment of a Centre of Excellence in Disaster Research and Resilience Building in JNU for promoting higher education and research within the multi-disciplinary framework.

v. **Disaster Management guidelines** for different kinds of disasters have been prepared

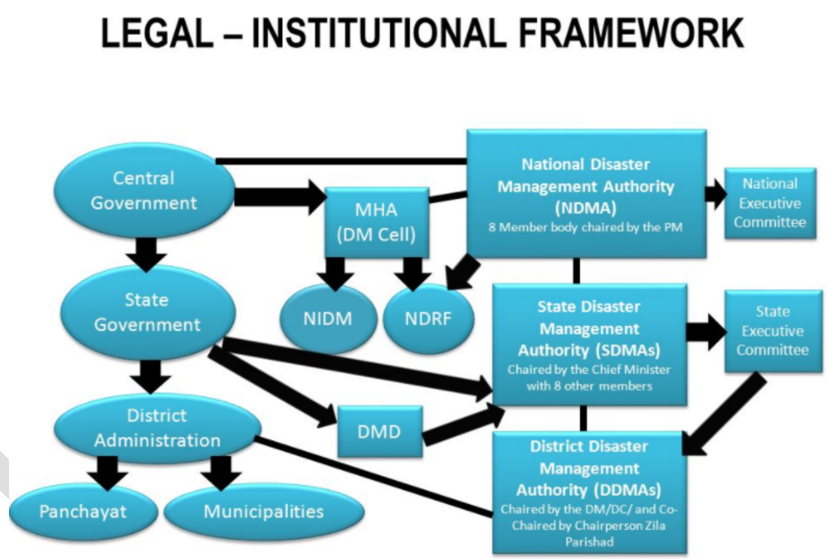
vi. India is also in the process of creating National level Database for Disasters in India to help fulfill Sendai Framework requirements.

- **Conclusion**: Class Discussion

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

7. INDIA'S INSTITUTIONAL FRAMWORK FOR DISASTER MANAGEMENT

- **Example Questions**
  - » "Ill-defined and conflicting institutional structures under the disaster management legal framework are among the major obstructions to effective disaster management" Discuss. [10 marks, 150 words]
- **Introduction:**
  - » The Constitution of India doesn't mention disaster management under any of the three lists under Schedule 7. Thus, it comes under the residuary powers of the Union. Therefore, power to legislate on the matter lies with the Parliament of India.
  - » After the 2004 Tsunami, the Indian Parliament passed the **Disaster Management Act, 2005** which provides for institutional framework to deal with disasters in India.



- **The Disaster Management Act,** 2005 has created **new institutions** at the National, state, district and local levels. The new institutional framework is as follows:

  i. **National Level**

  - » **National Disaster Management Authority (NDMA)** under chairmanship of the Prime Minister is responsible for policy formulation, making guidelines & best practices, and coordinating with SDMAs to ensure holistic and distributed approach to disaster management.
    - It also has the powers to approve National Plan and Plans of various ministries regarding DM
    - Powers to superintendence and control of NDRF.
  - ▪ **National Executive committee,** chaired by home secretary**,** is responsible for preparing national plan, assisting NDMA to discharge their functions; monitoring the implementation of the National Policy and ensuring compliance to the direction issued by the Central Government.
  - ▪ **National Institute for Disaster Management (NIDM)** is responsible for human resource development, training, research and development, documentation and database creation.
  - ▪ **National Disaster Response Force (NDRF)** is a specialized force working under the overall supervision of NDMA. It is responsible for quick response to a threatening disaster situation. It can also be deployed to provide assistance to civil authorities in case of impending disasters.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- In 2019, it also decided to train specialized unit in each battalion to help preserve monuments and other heritage structures battered by disasters.

ii. **State Level**
- SDMA, chaired by CM is responsible for policies and plans of DM at the state level. It also coordinates the implementation of state plan.
- **State Executive Committee** draws up the state DM plan as prescribed by the national as well as state authorities.

iii. **District Level**
- District Disaster Management Authority (DDMA)- chaired by DM and have elected representatives of the local authority as co-chairperson. It is responsible for planning, coordinating and implementing Disaster Management at district level.

iv. **Funding mechanism** under DMA, 2005 - Disaster Relief Fund and Disaster Mitigation Fund (at district, state and National Level)

- **National Policy on Disaster Management, 2009**
  » Approved in 2009
  » **Vision**: To build a safe and disaster resilient India by developing a holistic, proactive, multi-disaster oriented and technology driven strategy through a culture of prevention, mitigation, preparedness and response"

- **Analysis: Positives of the DMA, 2005**
  » The act rightly emphasizes the need to move from responding to disasters to effective preparedness, which has led to most states investing in resilient infrastructure, early warning systems and evacuation.
    - This has translated into timely warnings, relief shelters and massive evacuation exercises. All these have **reduced causalities**.
  » The **National Disaster Response Fund and State Disaster Response Funds** have helped in guiding immediate relief in the aftermath of disaster.

- **Some limitations associated with the Disaster Management Act**
  i. **Too Much Bureaucratization** -> top-down approach -> ignores the role of local communities and civil society organizations
  ii. **Poor Implementation ->** In Swaraj Abhiyan vs Union of India, 2016, the Supreme Court pulled government for poor and slow implementation of the law.
    - For e.g. the NDMF hasn't been created yet.
  iii. **Ill-defined and Conflicting institutional structures** under the disaster management legal framework
    - E.g. Overlap of role between NEC and already existing National Crisis Management Cell (NCMC).
  iv. **Need of clarity and transparency** on **how the political and administrative authorities respond to a tragedy**.
    - For e.g. **multiple nomenclature** (National disaster vs L3 Level Severity (NDM Guidelines) vs Calamity of Severe nature (DMD of Home Ministry)

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

v. **Neglected NDMA, NEC:** Vacancies, excessive executive incursion by MoHA, lack of regular meeting of NEC (reported in CAG report) etc.

vi. **Funding Misappropriation** (especially from SDRF)

vii. **Lack of Focus on Long Term Recovery**

- The DMA, 2005 largely focuses on improving preparedness, providing immediate relief, and protecting infrastructure. However, it neglects a key aspect of disaster management: Long term recovery.
- Post disaster relief and recovery **have been left to respective ministries and departments**.

- **Key Recommendations of ARC on NDMA, 2005**

i. **Decentralization:**

- **States should play primary role ->** Disaster Management should be the primary responsibility of state governments and union government should only play supportive role**.**
- **Local government's role should be brought to forefront** in disaster management.

ii. **Ending conflicting institutional structures:** 2nd ARC recommended the scrapping of NEC as its function coincides with that of the already existing National Crisis Management Committee (NCMC) chaired by Cabinet Secretary.

iii. **Standardize the methodology for assessing a Disaster:** The act should provide for the categorization of the disaster; this categorization will help in determining the level of authority primarily responsible for dealing with the disaster as well as the scale of response and relief.

iv. **Preventing misuse of the funds:** The act must provide for stringent punishments and penalties in case of misuse of funds for disaster management

- **Other Important Recommendations**

i. Place Disaster Management in the concurrent list to ensure vertical and horizontal linkages in effective disaster management.

ii. **Long term recovery** needs to be thought of alongside development in an integrated and comprehensive manner by combining with health, skill building, and livelihood diversification schemes.

- **Conclusion**

- There is a need to move towards a regime that exists on a unified, codified and systematic approach to disaster management in which duties of center and states are well defined.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

8. MANAGING RISKS OF DISASTER FOR SUSTAINABLE DEVELOPMENT

- **Example Questions**
  - » 'Disaster Risk reduction is the most crucial component of Crisis Management' Discuss [10 marks, 150 words]

- **Introduction**
  - » Over the years disaster management has **evolved from managing events of a disaster to managing the risks of a disaster**.
    - In the risk management approach to disaster, **various risks of hazards are analyzed**, and steps are taken to **reduce and control these risks**. Here both structural and non-structural steps can be taken for disaster risk reduction.
  - » **For the residual risk** which can't be prevented, there has to be **disaster preparedness** i.e., getting ready to respond to disaster effectively. It involves preparing for early warning, evacuation, search rescue etc. Preparedness further means having policies, strategies and resources in place for 'building back better' livelihoods, houses, infrastructure etc. devastated during the floods.

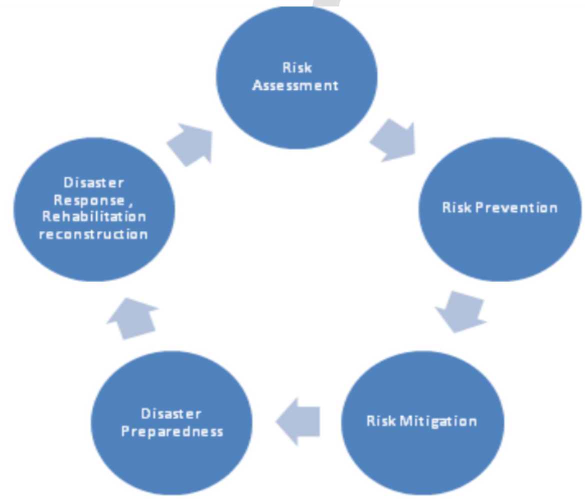


Figure 1 From Disaster Management to Disaster Risk Management

- **Significance of Disaster Risk Reduction**
  1. **Reduces the negative impact of disaster** on lives, livelihood and infrastructure
  2. **Sustainable Development**: Disasters can reduce the pace towards sustainable development, by damaging infrastructure, and diverting resources. By reducing its risks, we can increase our pace of movement towards sustainable development.
  3. **Economic advantages** - WB estimates that India has lost at least 2 percent of our GDP because of disaster. DRR measures can substantially reduce the economic losses.
  4. **Safeguarding Critical Infrastructure**: DRR can protect critical assets like Hospitals, schools, transportation networks which can be crucial for the proper functioning of the society.
  5. **DRR can fight poverty** as disaster generally impacts the vulnerable section the most.
  6. **Addressing Climate Change Impact**: DRR will include adapting to the changing climatic conditions, building resilience to extreme weather events, and mitigating the adverse impacts of climate change.
  7. **Promote International Cooperation:** Several disasters can transcend national boundaries and its risk reduction may require collaboration between multiple countries contributing to increased pressure towards international cooperation.

- **Disaster Resilient Sustainable Development got a new momentum in 2015** when three parallel yet inter-dependent processes converged to define the development agendas for the next one and a half decade and beyond.
  - » **Sendai Framework on DRR** 2015-2030

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- The **2030 Agenda for Sustainable Development** adopted by UNGA in Sep 2015
    - It embedded disaster risk management in as many as 8 out of 17 sustainable development goals.
- The **Paris Agreement on Climate Change** signed in Dec 2015 outlined 8 specific action areas for enhancing 'understanding, action and support for disaster risk reduction'.
    - These include Early Warning System; Emergency Preparedness; Slow onset events; Events that may involve irreversible and permanent loss and damage; Comprehensive risk assessment and management; Risk insurance facilities; climate risk pooling and other insurance solutions; Non-economic losses; Resilience of communities, livelihoods and ecosystems

- **Steps taken so far**
    - India has put in place legal and institutional mechanisms at various levels and deployed scientific and technological capabilities for disaster risk management with clearly visible impact on loss of lives, as was demonstrated during some of the recent meteorological disasters like cyclones.

- **Challenges**
    - **Disaster Preparedness and risk reduction** is invisible as compared to disaster response and **thus is sometimes ignored**.
        - Neither NDMA, nor various ministries and departments have come up with guidelines or concrete plan of action for building disaster resilience in their respective sectors.
    - **We have not been very successful** in dealing with
        - **Hydrological disasters** like floods or cloudbursts (Uttarakhand, Srinagar, Chennai, Kerala etc) or **geological disasters** like landslides (Malign and north Sikkim)
        - **Technological disasters** like road accidents or industrial accidents continue to spiral;
        - **Threats of biological disasters** like epidemics and pandemics loom large, while **environmental disasters** like depleting water resources and rising level of air pollution in rapidly growing urban settlements are causes of major concerns.
        - India's **ability to manage earthquakes** have not been tested yet since the 2001 Kutch earthquake.
    - **Our strong scientific base and traditional knowledge and understanding** of the natural and anthropogenic processes of risks of disaster are not being used in process of designing and implementation of social and economic development programmes, activities and projects, with the result that benefit of these projects for disaster risk reduction are not optimized and on the contrary some of these projects are directly or indirectly contributing to creation of new risks of disasters or exacerbation of existing risks of disasters.

- **Way Forward**
    - **Mainstreaming Disaster Risk Reduction:** Implementation of Sendai framework in conjunction with SDG goals, and Paris Climate Agreement provide an opportunities for addressing this hitherto neglected but challenging tasks of disaster risk management in India.
    - There is a need to **promote awareness generation** regarding adoption of disaster resilient building by laws, land use zoning, resource planning, establishment of early warning systems, and technical competence.
    - **Promote Knowledge sharing** among Disaster Management community

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- We need a common platform to create a versatile interface among policy-makers in the Government and disaster managers at all administrative levels.

- **Conclusion**
    - » If national targets for growth and development - including employment and trade - are to be realized, the shift from managing crisis to managing risk must be reflected in public policy frameworks and planning decision processes so as to enable risk informed investment and practices.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

9. FLOOD RELATED DISASTERS

1) FLOOD

- **Why in news?**
  - » Assam under grim flood situation (July 2024)

- **Past year Questions**
  - » Why are floods such a recurrent feature in India? Discuss the measures taken by the Government for flood control (1985, 20 marks)
  - » In what way can flood be converted into a sustainable source of irrigation and all-weather inland navigation in India. [2017, 250 words]

- **Introduction**
  - » Inundation of land and human settlements by the rise of water in the channels and its spill-over presents the condition of flooding. Flood is a natural disaster which affects some or the other part of the country almost every year now. (Kerala, Chennai, Assam, Bihar, UP etc.).
  - » According to **Asian Development Banks**, floods are the most devastating among climate related disasters in India. They account for more than 50% of all climate related disasters in the country.

- **Situation in India**
  - » In India, around 40 million hectares area is flood prone, which is 1/8th of the total area. As per the NDMA, the regions susceptible to flood lies mostly along Ganga and Brahmaputra basin. The coastal states of Odisha and Andhra Pradesh, parts of Telangana and Gujarat also witness yearly floods.
    - ▫ But the recent floods in Kerala and Tamil Nadu shows that even peninsular India is not more immune to flood disasters.

- **Causes of Floods**
  - » **Natural Causes:** Flood is generally seen as a natural phenomenon. It is associated with:
    - ▫ **Heavy Rainfall**
      - ▪ **Cyclones** (E.g. Cyclone Michuang in Chennai)
      - ▪ **Cloudburst**
      - ▪ **Monsoon Climat**e - all rainfall confined to a period
      - ▪ **Interaction of Western Disturbance (WD)** with the Monsoon Low Pressure system (cause of heavy rains in Punjab, J&K, Delhi in July 2023)
    - ▫ **Sediment Deposition**
      - ▪ Causes rivers to overflow or change paths

  - » **Manmade causes**: Experts believe that the recent increase in intensity of floods have to do a lot with human activities:
    - i. **Climate Change** has led to extreme variability in the intensity of rainfall which has increased the chances of floods.
      - ▪ For e.g., global warming has caused rainfall due to western disturbances even in Monsoon season in July 2023 causing huge rainfalls in NW India.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

ii. **Unplanned development along the natural drainage system** has led to rivers losing its buffer areas and thus any increase in the water levels is causing floods. This includes colonization of flood plains and river beds.
    - **E.g. - Situation of Yamuna**:
        - The process of physical demarcation of the Yamuna's floodplain - a basic step to help identify and protect the sensitive ecosystem from encroachment - is still incomplete (April 2024) despite court orders.
        - In 2015, the NGT had asked authorities to identify encroachments in the floodplains. It hasn't been done so far. In fact, illegal constructions continue to grow on the Yamuna flood plains.
        - Delhi Development Authority (DDA), in violation of NGT guidelines continues construction on crowded Yamuna flood plains in Delhi.

iii. **Indiscriminate Deforestation** has led to increased devastation due to floods. Trees generally acted as a breaker in the intensity of floods.
    - For e.g. According to Madhav Gadgil, if we would have protected Western Ghats, the loss and devastation by the Kerala floods of 2018 would have been less severe.

iv. **Unsustainable agri-practices** can also be considered an important factor behind the recent rise in floods.

v. **Inefficient Dam Management** sometimes lead to large scale release of water in small time period leading to flood conditions
    - E.g. Kerala floods pf 2018

vi. **Urban Floods** are also mostly a result of human made factors
    - **Blocking the natural flow of rivers**
    - **Destroying the natural sinks** like ponds, lakes etc.
    - **Concretization** - Reduces the seepage of water - all water flows and cause floods
    - **Improper Urban Planning** -> siltation of drainage system, Insufficient drainage system

- **Consequence of floods -** Life, Property, Infrastructure, Agriculture, Water Borne diseases etc.
    » According to Central Water Commission, the total flood related losses in the country were estimated to be over 37 lakh crore from 1953 to 2017.
    » As per the *the State of the Climate in Asia 2021* report, loss and damages from floods, storm cost India **$7.6 billion in 2021** alone.

- **Some positive impact**
    » It deposits fertile alluvial soil and thus perpetuates the fertility of the area.

- **Dealing with Flood Disasters/ Flood Management in India**
    a. Risk Reduction, Preparedness
    - **Flood Plane Zonation (FPZ)** to mitigate damages caused by floods and to allow rivers their 'Right to Way'. As a policy flood plain zonation has two major components: Removing Encroachment and Regulating Land Use.
        - **Remove Encroachment:** (Delineate Flood Plains by conducting survey; classify land according to risks; Relocate the exposed and provide compensation to them)

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- **Regulating Land Use:** Remove unauthorized construction; Regulate land use; Incorporate flood resilient design; Make public infrastructure flood resilient; Conserve wetlands (Prohibit reclamation, maintain existing wetland, recover the lost wetlands)

- **Other River Related Steps**
  - **Embankments:** e.g. Embankments on Yamuna in Delhi has been successful in controlling the flood to large extent.
  - **Periodic desilting of river**
  - **Watershed based master planning** and development legislated guidelines for each major river basin is needed.
    - It should demarcate ecologically sensitive zones. There must be clear land use plan for these zones specifying flood plains, protected forest areas, agricultural and plantation zones.
- **Afforestation:** the furry of flood could be minimized by planting trees in catchment areas of the river

- **Planned Scientific Development of Cities**
  - Protect natural sinks like Ponds, lakes etc., development away from the river channel, proper drainage infrastructure, regular cleaning of this infrastructure.
  - Review and revise **building by laws** to focus more on environmental sustainability. They should clearly provide that natural drainage and streams shall not be obstructed by this development/ building permit.

- **Continuous modernization of flood forecasting, early warning and decision support systems**
  - There is a need of more accurate rain forecast and more detailed warnings in place of the current categorization as "heavy" or "very heavy".
  - IMD needs more Doppler weather Radars which can extend the lead time of forecast by three days.
    - E.g. **IFLOWs-Mumbai** was launched in June 2020 as an state of art integrated flood Early Warning system for Mumbai to enhance the resilience of Mumbai specially during high rainfall events and cyclones.

- **Improving awareness and preparedness of all stakeholders** in the flood prone areas.
  - **Regular Drills in Flood Prone Areas** to ensure preparedness of NDRF and awareness among masses regarding steps to be taken during floods.
  - Introducing **capacity development interventions** for effective Flood Management (including education, training, capacity building, R&D, documentation) etc.
- **International Cooperation** with neighboring countries on flood controls as a number of rivers which cause flood in India originate from other neighboring countries.
  - For e.g. Dams on Rivers in Nepal can play an important role in controlling floods in the state of Bihar.

b. Response

- Improve the response system of NDRF especially for rural states like Bihar and Odisha.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- Need to **<u>enhance capacity building for catastrophic weather events</u>**
  - Serious attention needs to be given to <u>fast tracking the setting up of relief camps</u>, <u>crisis proof health infrastructure</u> and <u>stockpiling of dry ration and medicines</u>.
- Increased use of technologies <u>like drones to identify people</u> who are trapped in flood

c. Recovery

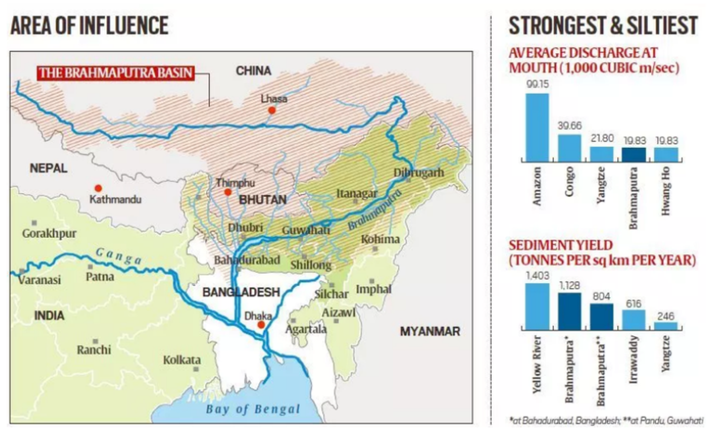
- <u>Special Focus on Water borne diseases</u> as they are the biggest killer in the post flood situation.
- Ensure that the <u>new infrastructure created</u> is resistant to floods.
- Bring in changes like broadening ecologically sensitive domain to protect more area from environmental degradation.

- **Conclusion:**
  » By recognizing the increasing threat of extreme precipitation and implementing proactive measures, India can improve its resilience to extreme weather events.

- **Conclusion**
  » India being a sub-tropical country with Monsoon kind of climate will remain vulnerable to floods due to heavy rainfall and increased climate variability. An efficient disaster management mechanism will ensure that these floods remain a natural phenomenon and doesn't become a natural disaster

2) CASE STUDY OF BRAHMAPUTRA

- **Why in news?**
  » Most of the districts of Assam face flood every year



- **Why is Assam so vulnerable to floods?**
  » **Incessant Monsoon Rainfall**
  » **Nature of River Brahmaputra** -> Dynamic and Unstable
    - It figures amongst the <u>world's top five rivers in terms of discharge</u> as well as the <u>sediments</u> that it brings.
    - Because of <u>earthquake prone nature</u> of the region, the river has not been able to acquire a <u>stable character</u>.
  » **Man-Made Factors** (Habitation, deforestation, increased population, destruction of wetlands)
    - For e.g. Sylhet districts traditional wetlands called 'haor' used to act as sponges absorbing runoff, have been destroyed, thereby <u>enabling the recent floods</u>.

- **How has government tried to address the factors that cause flood in Assam**

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- In its master plan on the river in 1982, the **Brahmaputra Board** has suggested that **dams and reservoirs** be built to mitigate floods.
    - But, the idea has been seen as a double edged sword.
    - Further, opposition from locals, and environmentalists on the grounds of displacement and destruction to ecology, prevented the plan from moving forward.
- **Building Embankments** were proposed as an **interim measure** to deal with floods and government has recently used this as the only major approach to control floods.
    - But since, there were temporary measures, most embankments which were built in 1980s were not strong enough and thus easily give way in case of overflow of rivers.
- **Dredging** is another thing that government has considered, but experts have opposed it as Brahmaputra sediment yield is amongst highest in the world and next year sediments will reverse the efforts of this dredging. So, even this hasn't been done.

- **Conclusion**
    - The above initiatives are clearly not the sustainable solution to floods. There is a need of a **basin wide approach** to the problem. There is a need of integrated basin management system that should ideally bring in all basin-sharing countries on board.

3) URBAN FLOOD

- **Why in news?**
    - The City of Delhi was flooded after the first rain of Monsoon 2024
    - Chennai has seen repeated urban Floods (in Dec 2015, Dec 2023); Similarly, Urban floods in Bengaluru in Sep 2022 was caused by heavy rain, poor drainage and worse infra.

- **Introduction**
    - Recent instances of floods such as the one in Delhi in June 2024, Chennai in Dec 2024, Bengaluru in Sep 2022, Chennai in 2015 and Mumbai in 2005 illustrate the **increasing vulnerability of Indian cities to this disaster**. A complex set of factors have worked together to deteriorate the condition of our cities and increase their susceptibility to this devastation.

- **Main Causes of Urban Floods**
    i. **Unplanned Urban Development:**
        - **Increasing Concretization of city** land reduced the seepage of water in the ground and has increased the runoff: NDMA Report
        - **Loss of Natural flood storage** in urban areas by filling of ponds and lakes to reclaim land for development.
            - For instance, in Chennai the number of water bodies have come down to less than 50 from 600 in 1980s. This became a major cause of 2015 floods.
            - Similarly, Bengaluru had 1,452 water bodies in 19th century, this has now reduced to only 193 lakes.
                - Here about 10,787 acres of lakes worth Rs 1.5 lakh crore has been encroached upon.
        - **Encroachment of Flood Plains** of the rivers have led to loss of natural flood storage.
            - 2013 Uttarakhand floods, Delhi 2023 floods.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

» **Rapid urbanization** has led to massive changes in land use patterns, as residential areas had sprung up in farmlands.

ii. **Improper and Inadequate Drainage system**

- A lot of sewerage and drainage network is old and lack volume to carry flood water.
  - For e.g. The current drainage system of Delhi is based on the 1976 master plan.
- Poor Desilting and blockage of drainage systems. This was the main reason for 2005 Mumbai floods and a major factor in Sep 2022 Bengaluru floods.
  - Improper Waste Management leads to a lot of solid waste blocking the drains. This hinders the flow of water during rainfall and contributes to floods.

iii. **Global Climate Change** have led to change in weather pattern which is sometimes causing unusually heavy rainfall thus causing floods in urban areas.

- **For e.g. On 5th Sep 2022, Bengaluru received 131.3 mm of rainfall.**

iv. **Social and Political apathy**

- Religious practices such as dumping of religious symbols, dead bodies etc. in rivers also lead to blocking of rivers. Inefficient management of gathering like Kumbha contribute to unnecessary concretization and thus floods.
- Socio-economic factors contribute to illegal encroachment of flood plains by slums etc. which increases the intensity of the urban floods.
- At the same time we have seen an absence of political will to give priority to the issue. This has happened both at national and international level.
  - This lack of political will has resulted into paucity of funds which delays the key drainage infrastructure
  - The river water information sharing has remained a major issue between India-Bangladesh and India China.

- **Consequences**
  - » **Human and Infra Loss** - deaths and devastation; loss of telecommunication, road and railway lines; increased probability of disease epidemic
  - » **Economic Losses:** Other than economic losses because of destruction of infrastructure, floods result in **traffic jams, temporary closure of business, destruction of property etc.** which leads to loss of manhours, hindering of economic activities etc.
  - » **Environmental Pollution** Urban floods also lead to washing away of various pollutants including industrial waste into water bodies thus intensifying river pollution.

- **Way forward**
  - » **Promote the ideas of Sponge Cities -> Urban planning should keep in mind the geological and hydrological cycle:** Planned Development of cities should ensure that flood plains are not encroached upon, sinks like ponds are protected/restored and pavements are porous to allow infiltration of rainwater in the ground.
    - There should be increased focus on these goals through an **Mission on Sponge Cities**.
  - » **Depopulation the river flood plains**
  - » **Improvement of drainage system.**
    - Proper maintenance, desilting of existing drainage system

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- Providing alternative drainage path for flood waters (may be underground)
- Control of solid waste entering the drainage systems through proper Solid Waste Management

» **Change in social attitude of Common Citizen** will go a long way in controlling urban floods

- Reduction of solid waste, promoting environment friendly religious practices can all contribute towards limiting urban floods.

» **Disaster preparedness:** Even after all proper steps, nature may cause havoc and cause floods, therefore a **proper disaster management plan** should be prepared by the ULBs to be battle ready in emergency situations. Fresh Hazard profiles should be created for the cities based on the historic as well as recent flood vulnerabilities.

- **Conclusion**
  » We must not allow nature, human conduct, and urbanisation to be mystified and rendered as trans-historic villains. We can learn to live with nature, we can regulate human conduct through the state, and we can strategically design where we build. We need to urgently rebuild our cities such that they have the sponginess to absorb and release water without causing so much misery and so much damage to the most vulnerable of our citizens, as we have seen.

4) CLOUD BURST

- **Practice Questions:**
  » Explain the mechanism and occurrence of cloudburst in the context of the Indian subcontinent. Discuss two recent examples. [Mains 2022, 10 marks, 150 words]
  » Cloudbursts are often associated with flash floods. Explain the relationship between cloudbursts and flash floods and discuss the challenges in managing flash flood events. [10 marks, 150 words]

- **What is cloudburst?**
  » A cloudburst refers to an **extreme amount of rain that happens in a short period**, sometimes accompanied by hail and thunder. IMD defines it as unexpected precipitation **exceeding 10 cm per hour** over a geographical region of approximately 20-30 sq km.
    - For e.g. the 2013 floods in Kedarnath were caused by Cloud Burst. In 2021, Amarnath region was impacted by cloudburst.
  » **Impact**: This sudden discharge of rain leads to floods including flash floods, landslides etc. which may result into human casualties, and property loss.

- **Mechanism: How does cloudburst occur?**
  » When cumulonimbus clouds (which stretch to even 13-14 kms in height) are trapped over a region or there is no air movement for them to disperse, they discharge over a specific area.
    - Here, saturated clouds ready to condense into rain can't produce rain, due to the upward movement of the very warm current of air.
    - Instead of falling downwards, raindrops are carried upwards by the air current. New drops are formed and existing raindrops increase in size. After a point, the **raindrop are too heavy for the cloud to hold on to**, and they drop down together in a quick flash

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- **Other key aspects:**
    - It is very **difficult to forecast** the event due to its very small scale in space and time.
        - To monitor or nowcast (forecasting few hours of lead time) the cloudburst, we need to have **dense radar network** over the cloudburst-prone areas or one need to have a v**ery high resolution weather forecasting models** to resolve the scale of cloudburst. Doppler radar can be very useful in predicting them.
    - **Mountain regions are more prone** to cloudburst due to **orography (terrain and elevation)**, though they may occur in plains as well.

- **Way forward:**
    » **Hazard zonation mapping**: Identifying the areas vulnerable to flash floods.
    » **Building flood resistant infrastructure**: To reduce damages due to flash floods
    » **Regulating settlements** in the river banks
    » **Improving forecasting (nowcasting) Infrastructure**: Increasing the coverage of doppler radars. Currently Himalayan region has 7 doppler radars (2 each in J&K and Uttarakhand, 1 each in Assam, Meghalaya and Tripura.
    » **Strengthening institutions** to provide quick response at the time of cloudburst in the form of emergency evacuation, medicine etc.

- **Conclusion:**
    » Bay taking steps to predict, prepare, and respond to these events, we can reduce the loss of life and the property damage that they cause.

5) GLACIAL LAKE OUTBURST FLOOD

- **Why in news?**
    » Why Uttarakhand government wants to evaluate the risk of Glacial Lake Outburst Floods (April 2024: Source-IE)

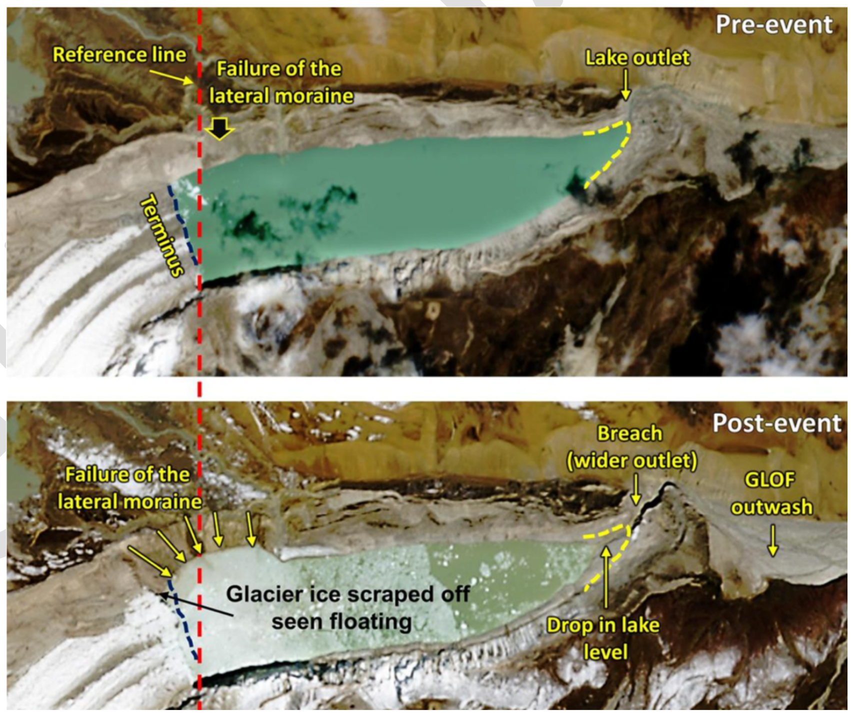
- **Example Questions**
    » Discuss the key factors which is making Himalayan region more vulnerable to Glacial Lake Outburst floods (GLOF). In light of the recent NDMA guidelines, suggest measures to reduce risks of GLOF disasters (15 marks, 250 words)

- **Introduction**
    » GLOFs are sudden fast flowing release of glacial lake water that move downslopes as a result of dam failure. They are recognized in the National Disaster Management Plan (NDMP) 2019 of India as a **potential climatological disaster**.
- **Current Situation**:
    » A study, 'Glacial lake outburst floods threaten million globally' published in the journal **Nature** in Feb 2023 highlights that:
        - **Around 15 million people globally** face the risk of GLOF.
        - Around 20% of them (**3 million)** live in India.
        - India, Pakistan, Peru and China have more than 50% of the vulnerable people.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- **Causes of increasing GLOF**
    - » **Global Warming -> Climate Change**
        - ▫ **Increasing number of Glacial Lakes due to** acceleration of glacier melt in recent decades.
        - ▫ **Increased water pressure** due to more water being available due to Global Warming.
        - ▫ For e.g. As per ISRO's study through satellite based analysis, of the 2,461 lakes larger than 10 hectares in the catchment of Indian Himalayan River basins, **676** lakes have notably expanded since 1984.
    - » **Ice or rock avalanches, Erosions or other natural disruptions**
    - » **Earthquakes** - Himalayan region is especially prone to earthquakes
    - » **Human Activities** -> increased tourism, expansion of roads and hydropower projects, deforestation etc have also increased the vulnerability of burst in these lacks.

- **Recent Examples:**
    - » **Flash Floods in Sikkim in Oct 2023** which killed around 100 people, destroyed infrastructure like bridges and roads, and damaged state's largest hydropower project, the **1.2 GW Teesta-III**.
        - ▪ Cause: Access rainfall and GLOF of South Lhonak Lake. It was already recognized as potentially hazardous and scientists at the National Remote Sensing Centre had warned of a 42% chance of GLOF in as early as 2013.
        - ▪ **How it happened in Oct 2023** (See image)



    - ▫ The **Chamoli Flash floods of 2021** may have caused economic damages worth Rs 4,000 crore. It swept away the Rishiganga Hydel Power Project and inflicted substantial damage on the Tapovan Power Project.
    - ▫ **2013 Kedarnath** flash floods was also result of GLOF.

- **Adverse Impact**
    - ▫ These floods pose **severe geomorphological hazards and risks**
        - ▪ It can wreak havoc on all man made structures located along the path and thus endanger people, infrastructure, fields and livestock.
    - ▫ **Long term Climate Impact** may be caused by large glacial lake as they would increase the amount of water in ocean and reduce it in Himalayas.

- **Steps taken so far:**

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- **NDMA** (under Home Ministry), has identified 188 glacial lakes in the Himalayan states that can potentially breach because of heavy rainfall.
- NDMA has also prepared **Guidelines on the Management of Glacial Lake Outburst Floods (GLOFs)** which are aimed at improving the administrative response; drawing on international best practices; and brining together the relevant scientific capabilities of the nation.

- **Key Highlight of the NDMA Guidelines**
  i. **Inventorization: Hazard and Risk Mapping**
    - **Regular monitoring** of glacial lakes using satellite observations.
    - **Cooperation with neighbouring countries** (Nepal, Bhutan and China) to identify transboundary threats and manage it properly**.**
  ii. **Reduction of Hazards**
    - **Short term actions** - lowering the lake level through siphoning
      - For instance, high density PVC pipes were installed in **South Lhonak lake in Sikki**m, to reduce the pressure on the lake
    - **Long Term Actions**
      - **Artificial drainage channels** to lower lake levels
      - Reinforcement of dam
      - Enhancement of river cross section/ protection from erosion
    - **Restricting constructions and development** in GLOF prone areas is a very efficient means to reduce risks at no cost.
    - **Develop regulation for Land Use Planning** in GLOF areas.
  iii. **Reduction of Exposure**
    - Establishment of **Early Warning System.:** Comprehensive alarm system - including classical alarming infrastructure as well as modern technology using smart phone notifications etc.
    - **Evacuation** based on EWS
    - **Involve local population closely** from the beginning in the design, planning and implementation of risk reduction and management strategies in a transparent collaboration mechanism.
    - **Awareness and Preparednes**s through posters, social media, apps etc.
  iv. **Capacity Development** -
    - Apart from specialized forces such as **NDRF**, **ITBP**, and the **ARMY,** the guidelines emphasize on need for **trained local manpower**.
    - Training of professionals and practitioners;
    - **Heavy earthmoving and search and rescue equipment**, as well as motor launches, country boats, inflatable rubber boats, life jackets etc.
    - Setting up **Quick Reaction Medical Teams**, **mobile field hospitals**, **Accident Relief Medical Vans**, and **heli-ambulances** in areas inaccessible by roads.
  v. **Promote R&D in GLOF Management**
    - Strengthening Academic Education in relevant disciplines from natural and social sciences.
    - Promote development of **Modelling tools** to simulate the entire chain of mass movement and outburst process
    - **Historical records** should be effectively used to understand flood processes.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- Expand the use of **local knowledge, experience of local peopl**e. Engaging the local population in **joint-knowledge production** is considered indispensable for effective community based disaster risk management.

vi. **Regulation and Enforcement**
- **A well drafted techno-legal** regime is necessary to prevent future development of GLOF and protect existing Glaciers.
- The regime should include a Himalaya GLOF mitigation Policy, no habitation and construction zones; and provisions for strict implementation.

- **Other steps which needs to be taken**:
  - **Institutional Improvement:** Need of a **nodal agency** to coordinate all the researches related to glaciers in the region .
  - **Sustainable Development: Restricting Tourism; Effective EIA** for new projects.
  - **International Cooperation**: GLOF risk is transboundary in nature, thus there is an urgent need for a comprehensive regional risk governance framework including India, Nepal, Bhutan etc.
  - **Fighting Climate Change**: Eventually, a major core reason for GLOF is CC and rising sizes of the lakes. This in long term can only be solved if the global community can work together to achieve Paris Climate Change Targets.

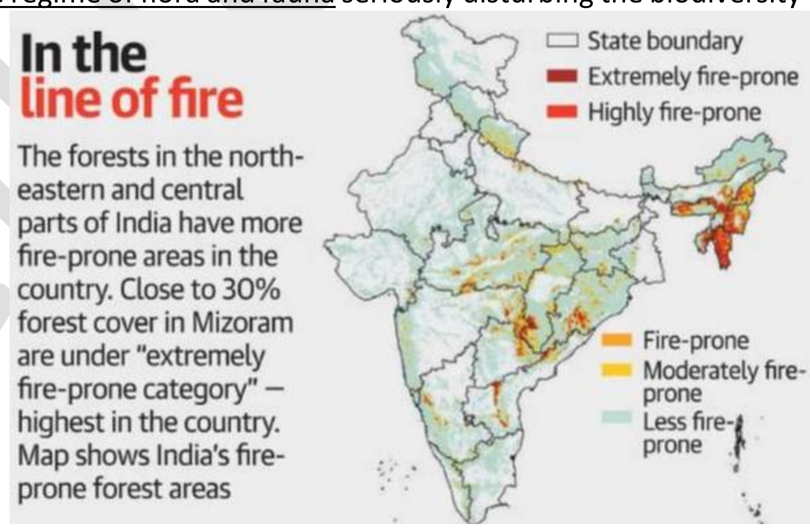
2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

10. FOREST FIRES

- **Why in news?**
    - » Despite being a significant source of greenhouse gases, wildfire emissions remain underestimated (July 2024; Source DTE)
    - » As per forest department, between January and June 2024 there have been 1,309 forest fires in Uttarakhand - up from 733 in the entire year in 2023.

- **Example Questions**
    i. Discuss the various causes and consequences of forest fires. What are the key features of National Master Plan for Forest Fires. [15 marks, 250 words]
    ii. Discuss the role played by climate change in aggravating the Forest Fire situation in the country. [10 marks, 150 words]
    iii. Discuss the temporal and spatial variation of fire vulnerability of Indian forests. Why have cases of forest fires increased in recent years? [10 marks, 150 words]

- **Introduction:**
    - » Fires have played a very important role in shaping forests since time immemorial. Foresters use it as a tool in scientific forest management as certain species regenerate and establish well under light fire conditions.
    - » However, uncontrolled fires is a major cause of degradation of forests. It **poses threat** not only to forest wealth, but also to entire regime of flora and fauna seriously disturbing the biodiversity and ecology and environment of the region.



- **Situation in India**
    - » According to Indian Forest Survey, 2019 (a report by Forest Survey of India), about **21.4% of forest cover in India** is prone to fires (extremely, very highly and highly fire prone), with forest in north-eastern region and Central India being the most vulnerable.

- **The most vulnerable regions** are **northeast, Central India** and **Himalayan Belt of Uttarakhand & Himachal Pradesh**
    - » **Northeast:** 7 states of northeast account for 33% of total alerts in India.
        - **Reason: Slash and burn agriculture/ jhoom cultivation** causes forest fire here. Otherwise, the tropical evergreen forests are not likely to catch fire easily.
    - » **Central India: Dry deciduous forest** of central India catches fire easily.
    - » **Himalayan belt** is prone to such fires, especially when there is **less rain in pre-monsoon period**. The **Pine Forest in Garhwal and Kumaon hills** are susceptible to forest fires more than any other kind of forest.
- **Temporal Distribution**

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

» The vulnerability is highest between **Feb and mid-June** period. This is the period of **hot and dry weather** when the soil moisture is at its lowest.

- **Main Causes: Natural and Manmade**
- **Natural Causes**
    » Environmental conditions such as temperature, wind speed and directions, level of moisture in soil and atmosphere and duration of dry spell. **El-Nino** which affects western disturbances, reducing pre-monsoon rains cause warmer dryer climate and thus make our forests more vulnerable to forest fires

    » In very dry season events like **lightening**, volcanic eruption, sparks from rock falls etc. can cause spontaneous combustion.

    » Other natural cause is friction of bamboos swaying in high wind velocity.

- **Human Related Causes**: Forest Survey of India's technical study says that **over 95% of fire incidents in India are anthropogenic in nature**. They can be intentional or unintentional
    » **Climate Change** is increasing the vulnerability of forests to fire in the country. More **heatwaves** in summers are leading to more forest fires. Fires may start by other causes
    - The fires are further stoking further heating leading to **a deadly cycle**.

    » **Grazers and gatherers** of various forest products starting small fires to obtain good grazing grass as well as to facilitate gathering of minor forest produce like flowers of Madhuca Indica and leaves of Diospyros melanoxylon.

    » **Use of Fire** to **ward off wild animals**, for **recreation, discarded cigarette buds**.

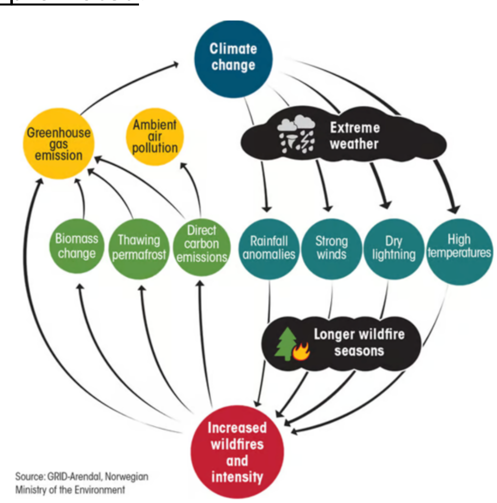
    » **Shifting Cultivation/Jhum Cultivation** (especially in NE region of India and parts of Andhra and Orissa) is a major cause.

    » **Arson** (e.g. by Timber Mafia), sparks from equipment, **power line arcs etc**.

- **Effects of Forest Fires**
    » **Economic Loss:** In India, close to 1.7 lakh villages are located in the proximity of forests; the livelihood of around 200 million people in India is dependent on fuelwood, bamboo, fodder, and small timber.

    » **Environmental Loss:**
    ▫ Loss of biodiversity and extinction of plants and animals.
        ▪ Fire resistant plants are disproportionately promoted.
        ▪ The area may also become vulnerable to increased penetration of **invasive species**.
    ▫ Loss of natural regeneration and reduction in forest cover
    ▫ Source of pollutants including black carbon.
    ▫ Degradation in soil quality and soil erosion impact the long-term productivity.

    » **Climate Change:** As per EU's Copernicus Atmosphere Monitoring Services (CAMS), in 2023, wildfires globally released **7,330 million tonnes** of CO2, which is only lower than China's emissions (and higher than emission by USA).



Source: GRID-Arendal, Norwegian Ministry of the Environment

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

- Climate change further leads to more intensive wildfire making it a loop.
- **Change in the microclimate of the area** with unhealthy living conditions
- Increase in **man-wildlife conflict**

- **Forest fires are difficult to control**
  - **Locality of the forest and access to it** -> hurdles in firefighting efforts
    - Difficult to transport heavy vehicles loaded with water into this forests.
  - **Shortage of staff** -> specially during peak season.

- **Role of Forest Survey of India (FSI) in Forest Fire Monitoring**
  - FSI has been assisting the forestry professionals in the country in many ways to deal with the problems associated with forest fires by using the latest remote sensing and communication technology.
  - **Following are some of FSI's initiatives:**
    1. **Near Real Time (NRT) Forest Fire Alerts**
       - FSI **monitors** forest fire events through satellites on two platforms - **MODIS and SNPP-VIIRS,** both in collaboration with the US NASA and ISRO.
       - **Forest Fire Alert 2.0** is the revamped alert system that was launched in Jan 2017.
    2. **Forest Fire Pre-Warning Alerts**
       - The alerts to state forest departments are based on **parameters like Forest cover, Forest type, Climatic Variables** (Temperature and Rainfall) and recent fire incidences over the area.
    3. **Burnt Scar Studies**
       - Assess the forest area affected by the forest fires to assess the damage to forest and biodiversity as well as to plan restoration measures.

- In **2019 FSI report**, **Fire prone forest areas have been identified for the first time on a country-wide level** to help state forest departments make suitable strategies to fight forest fires.

- **Other steps taken by India to tackle Forest Fires**
  - **National Action Plan on Forest Fires** (NAPFF), 2018
    - Focuses on **forest communities** - enable them, engage them and empower them to work with State Forest Department
    - Focus on **capacity building** of forest personnel and institutions.
  - **Forest Fire Prevention and Management Scheme**
    - Centrally sponsored program to assist states in dealing with forest fires
  - **Initiatives to fight climate change**
    - NAPCC
    - Renewable Energy Targets
- **Way Forward: The needs of fire management**
  - The Ministry of Environment and Forests, Government of India, has prepared a **National Master Plan for Forest Fire** Control. This plan proposes to introduce a well-coordinated and integrated fire-management program that includes the following components:
    1. **Prevention of human-caused fire**s through education and environmental modification.
       - It will include silvicultural activities, engineering works, people participation, and education and enforcement.

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

ii. **Prompt detection of fires** through a well-coordinated network of observation points, efficient ground patrolling, and communication networks. Remote sensing technology is to be given due importance in fire detection. For successful fire management and administration, a National Fire Danger Rating System (NFDRS) and Fire Forecasting System are to be developed in the country.
iii. **Fast initial attack measures**; **Vigorous follow up action**.
iv. **Harness local expertise**: Fire management **should focus on prevention and not suppression** and **tribal communities with traditional knowledge** should be roped in for effective management. In case of fire event, it is the locals who are first responders and contribute to nipping the problem in its bud.
v. Introducing a **forest fuel modification system** at strategic points.
vi. **Increasing Capacity:**
    - **Forest department's firefighting capabilities** also needs to be strengthened both in terms of human resource and **technical know-how**.
    - **Special emphasis** on **research, training and development** needs to be there

- In long run, the global community will also need to move **more aggressively towards achieving Paris Climate goals**.

- **Conclusion**
    - In an era of climate crisis and human encroachment into forest areas, it's time to learn form the crises of the US and Australia and bolster the fire fighting infrastructure.

11. NEXT CLASS

- Tsunami, Earthquake, Landslide, Heatwaves, Droughts, Dam Safety, Biological Disaster, Oil Spills, Building Collapse, Building Fires, Industrial accidents

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005
Ph: 08045248491, 07041021151 | Email: students@levelupias.com

GS FOUNDATION UPSC CSE 2025

LevelUp IAS

Lectures | Prelims-Mains Tests | Current Affairs | CSAT | Essay | Personalized Mentorship

Level 'O' → 01 Oct → 02 Feb → 03 Prelims → 04 Mains → 05 Interview → LBSNAA

Building a solid foundation
- Focus on fundamentals
- Big 4: History, Polity, Geography, Economics
- Current Affairs Phase -1
- Answer Writing structure & basics

Gaining momentum
- Interlinkages with GS topics
- Ethics, Essay, IR, Internal Security, Governance, Society, Disaster Management, etc.
- CA Phase 1

Intensive Prelims Mode
- Target Prelims Program
- Current Affairs Ph - 2
- Prelims Test Series
- CSAT Module

Mains Mode
- Target Mains
- Focus on GS 2 and GS 3 under Current Affairs Ph -3
- Answer Writing and consolidation

Interview Guidance
- Personality assessment and evaluation
- Mocks
- DAF based preparation
- 1-1 sessions and more

LevelUp HALL of FAME

NEW BATCH STARTS
24th JUNE 2024

2nd Floor, 45 Pusa Road, Opp. Metro Pillar 128, Karol Bagh, New Delhi-110005 | 08045248491 7041021151